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Microprocessor and Embedded Systems Laboratory

Final Project Report

Section: C1 Group: 08

IOT Based Real Time Intravenous Fluid Monitoring & Flow Control System

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Academic Honesty Statement:

"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

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1 Abstract

This project presents the design and development of an IoT-enabled Intravenous (IV) Fluid Monitoring and Control System for enhanced patient safety and healthcare efficiency. The system integrates real-time monitoring of IV fluid levels with the ability to regulate the flow automatically or allow manual adjustment by medical staff through a connected IoT interface. Additionally, the system is equipped with biomedical sensors to track vital signs such as body temperature and pulse rate, enabling it to detect abnormalities like fever or irregular heartbeat. Upon detecting critical conditions—such as low fluid levels, flow rate deviations, or physiological anomalies—the system sends immediate alerts to a centralized application accessible by doctors and nurses. This proactive notification mechanism ensures timely intervention, reduces the risk of human error, and enhances the overall quality of patient care.

2 Introduction

Intravenous (IV) therapy is a critical component of modern healthcare, where continuous fluid administration must be carefully monitored to ensure patient safety. Traditional manual monitoring of IV fluid levels is prone to human error, potentially leading to severe complications such as blood backflow, infection, and vein damage. This project proposes an IoT-based real-time intravenous fluid monitoring and flow control system that integrates sensing, control, and communication technologies. Using load cell or laser sensors, the system continuously tracks fluid levels, while a servo motor-controlled valve automatically adjusts the flow as needed. The system also monitors patient anomalies like fever and irregular pulse. Real-time data is transmitted via Wi-Fi to an IoT platform (Blynk), enabling healthcare providers to remotely monitor and respond to any abnormalities. The system aims to enhance patient safety, reduce nurse workload, and improve healthcare efficiency through automation and remote monitoring.

With the rapid advancement of Internet of Things (IoT) technology, there is a growing opportunity to enhance healthcare systems through automation, remote monitoring, and intelligent alert mechanisms. This project proposes an IoT-enabled IV fluid monitoring and control system that addresses the limitations of traditional methods. By incorporating sensors and microcontrollers, the system continuously monitors IV fluid levels and flow rates, while also tracking vital signs like body temperature and pulse rate. In case of abnormalities—such as fluid depletion, flow irregularities, or signs of fever or arrhythmia—the system automatically triggers alerts through a connected application, allowing timely response from medical professionals.

The system is also capable of automatically regulating the IV flow rate or enabling manual adjustments via a secure online interface, ensuring precise fluid delivery tailored to the patient’s needs. This smart healthcare solution aims to improve patient outcomes, reduce the workload on medical staff, and minimize human error through real-time, data-driven decision support.

3 Design

3.1 Problem Formulation (PO(b))

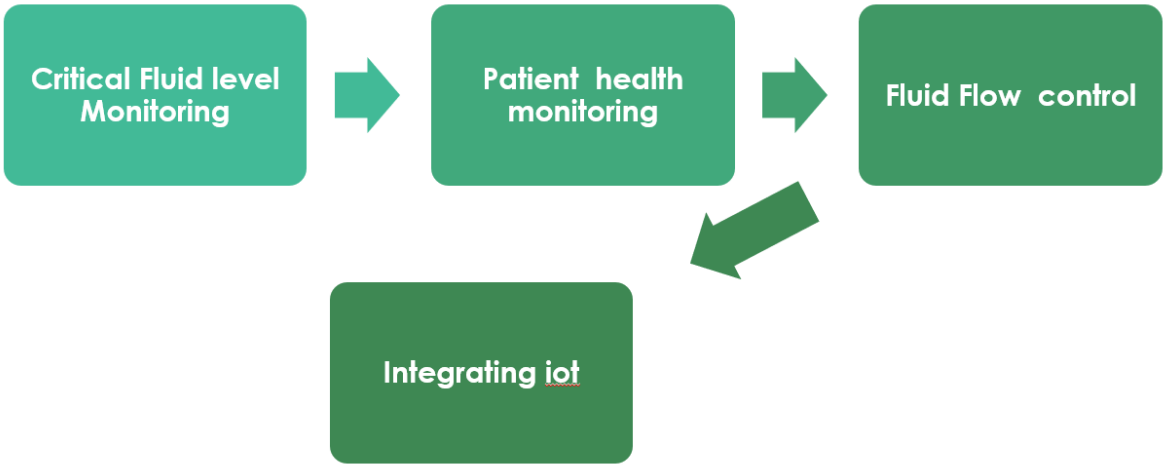
3.1.1 Identification of Scope

The scope of this project is to develop a smart, IoT-enabled system for real-time monitoring and control of intravenous (IV) fluid administration, with a focus on enhancing patient safety and reducing the burden on healthcare workers. The system will monitor fluid levels using either a load cell or a non-contact laser-photodiode module and send alerts when levels fall below a defined threshold. It will also track vital signs such as body temperature and pulse rate using low-power sensors to detect abnormalities like fever or irregular heartbeat. Upon detecting such conditions, the system will either automatically adjust the IV flow rate or allow remote intervention by doctors or nurses through a cloud-based interface using the Blynk platform. The ESP32 microcontroller will handle data collection and transmission via Wi-Fi, with secure, periodic updates to the cloud. A user-friendly interface will enable healthcare professionals to visualize patient data and control the system remotely. Designed for affordability, portability, and low power consumption, the system is suitable for both urban hospitals and under-resourced rural clinics, aiming to improve clinical outcomes and promote healthcare equity.

3.1.2 Literature Review

Our IV monitoring system and healthcare automation project reveals that conventional IV setups are predominantly manual and lack real-time sensing capabilities, posing risks in critical care scenarios. Many earlier smart IV solutions utilized high-cost sensors such as ultrasonic transducers or weight-based systems, limiting their accessibility and scalability. Studies on the integration of IoT in healthcare, particularly under the Internet of Medical Things (IoMT) framework, demonstrated significant benefits including reduced human error, improved monitoring accuracy, and faster emergency response times. Furthermore, the application of sensor fusion techniques—combining data from multiple sensors—has been shown to enhance the precision and reliability of biomedical monitoring systems. Based on these findings, our approach emphasizes the use of low-cost, non-contact laser-photodiode sensing, integration of vital sign monitoring for intelligent flow regulation, and the deployment of a cloud-based platform (Blynk) to enable real-time alerts and remote intervention, aligning with current trends in cost-effective and responsive healthcare automation.

3.1.3 Formulation of Problem



Critical Fluid Level Monitoring

Problem: In traditional IV systems, there is no continuous automated monitoring of fluid levels, which may result in the bottle running dry unnoticed—potentially causing air to enter the bloodstream.

Goal: Introduce a system that continuously tracks the fluid level and detects critical thresholds.

Patient Health Monitoring

Problem: Patients receiving IV fluids may also experience sudden changes in vital signs (e.g., fever or irregular pulse), but these are not monitored by basic IV setups.

Goal: Integrate vital sign sensors to monitor parameters like body temperature and pulse rate in real-time.

Fluid Flow Control

Problem: Manual adjustment of IV flow rate is error-prone and labor-intensive, especially in busy hospital settings.

Goal: Implement automated or remote-controlled flow regulation based on real-time health and fluid data.

Integrating IoT

Problem: There's no centralized, connected system for alerting staff or allowing remote access to monitor and control IV administration.

Goal: Use IoT technology to:

- Send alerts to nurses/doctors via a mobile or web app.
- Enable remote monitoring and control of the fluid delivery system.
- Log and visualize patient data for better diagnosis and follow-up..

This structured breakdown forms the foundation of the problem statement and guides the system design to solve critical safety and efficiency issues in intravenous fluid therapy.

3.1.4 Analysis

The system design integrates hardware components and IoT connectivity to build a smart IV fluid monitoring and control solution. The design is modular and divided into the following functional blocks:

1. Fluid Level Monitoring System

Components Used:

- Laser Module (HW493)
- Photodiode
- Potentiometer (Pot)

Working Principle:
A laser beam (HW493) is directed toward a photodiode sensor placed across the IV bottle. As the fluid level drops, the light path changes or gets interrupted, altering the signal received by the photodiode. The photodiode converts light intensity into voltage—low light (fluid blocking) vs. high light (no fluid). A potentiometer is used to calibrate the sensitivity (threshold) of the light received, allowing fine-tuning for different bottle sizes or laser strengths.
Purpose:
Detect low fluid levels in real time and alert the system when the fluid is about to run out.

2. Patient Vital Signs Monitoring

A. Body Temperature Monitoring

LM35 Temperature Sensor:
Analog sensor for measuring contact temperature (can be used near the skin). Provides linear output in mV/°C (10 mV per °C).

B. Heart Rate Monitoring

Heart Rate Pulse Sensor:
An optical sensor that detects blood volume changes through the skin .Usually placed on a fingertip or earlobe. Outputs analog pulses corresponding to heartbeat rate.
Purpose:
These sensors allow the system to track physiological abnormalities like fever and irregular pulse. The data is used for alert generation and dynamic IV flow adjustment.

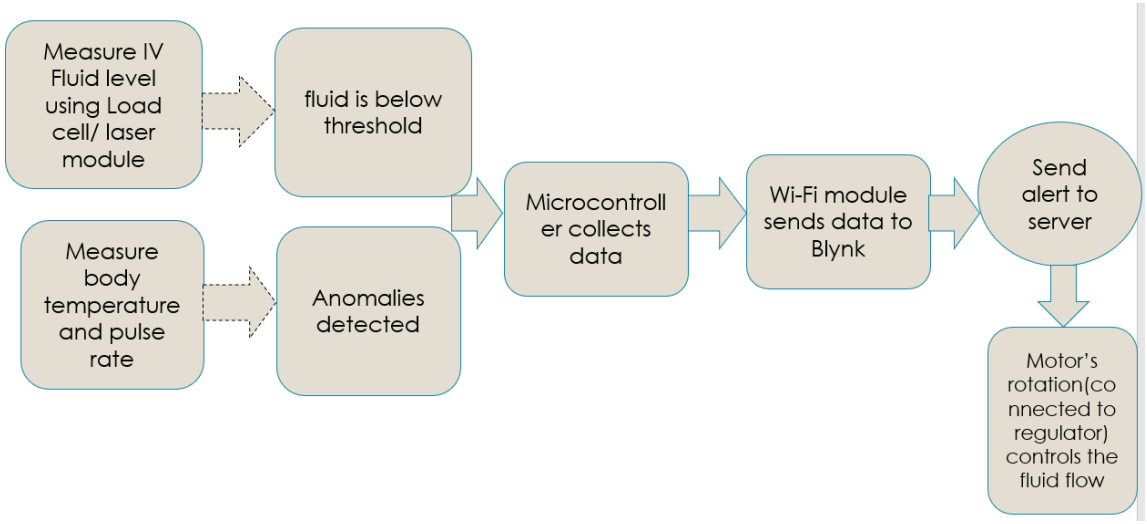
3. Fluid Flow Control Mechanism
Component Used:
Rack and Pinion Gear System
Working Principle: The rack and pinion system is mechanically connected to the IV roller clamp or flow regulator. A motorized setup moves the rack (linear movement) to rotate the pinion, which in turn adjusts the clamp to increase or decrease flow. This allows automated control of IV flow rate based on patient condition or fluid status.
Purpose: Adjust the IV fluid flow automatically (in emergency) or manually via IoT app interface.

4. IoT Integration and Cloud Communication
Component Used:
ESP32 Module
Function: Acts as the central processing unit and IoT controller. Collects sensor data (fluid level, temperature, pulse). Sends real-time data to the Blynk IoT platform via Wi-Fi. Receives remote commands from doctors/nurses to:

- ❖ Adjust flow rate.
- ❖ Turn alarms on/off.
- ❖ Monitor multiple patients remotely.

Purpose: Enables real-time monitoring, alerting, and control via smartphones or web dashboards. Stores and displays logs, graphs, and notification history.

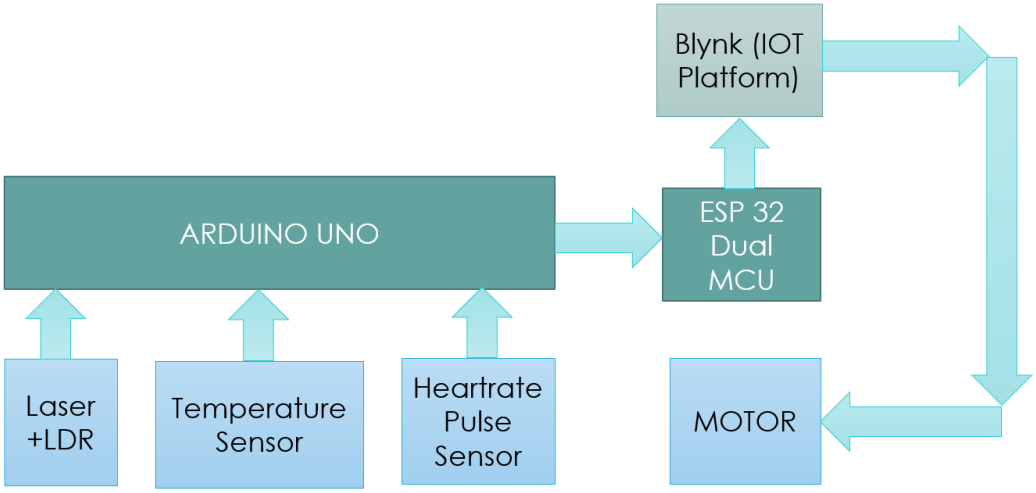
3.2 Design Method (PO(a))

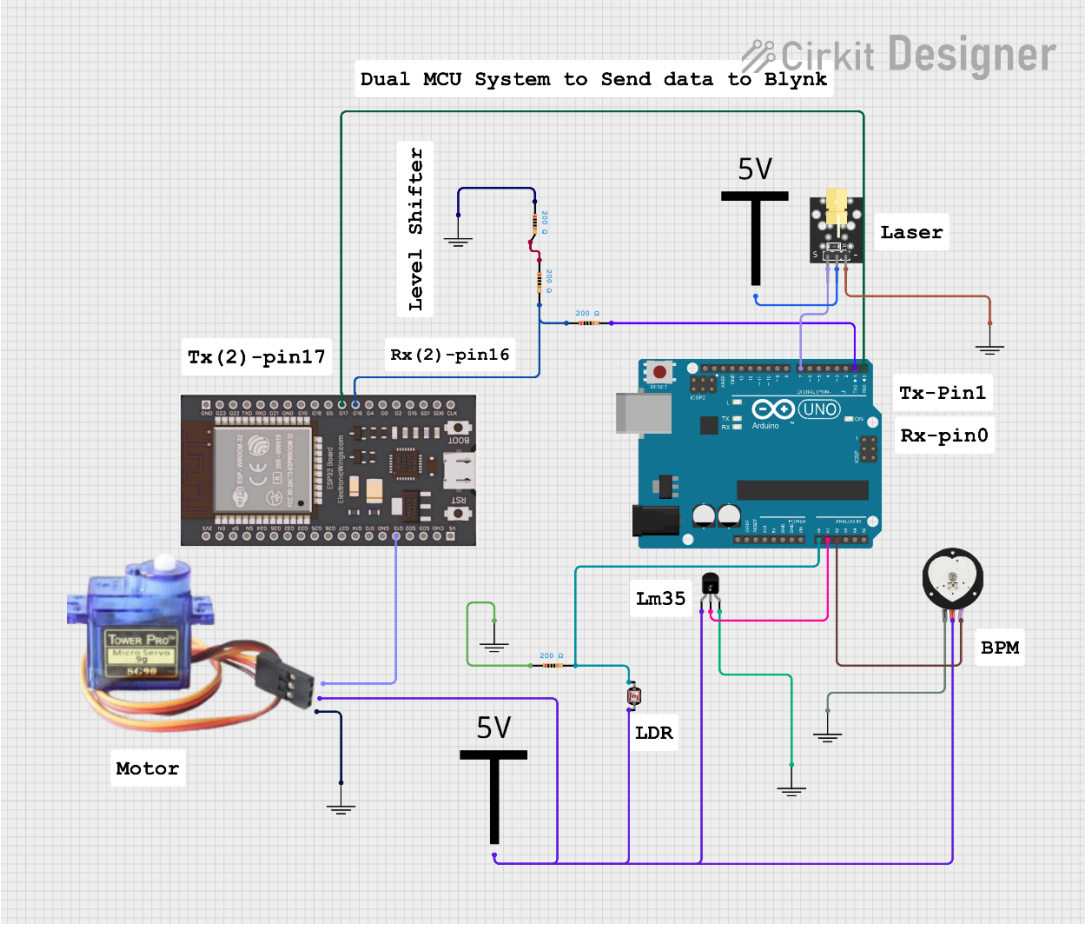


Function	Component(s) Used	Purpose
Fluid Level Monitoring	Laser (HW493), Photodiode, Potentiometer	Detect low fluid level via light interruption
Body Temperature Monitoring	LM35	Detect fever or abnormal temperature conditions
Pulse Monitoring	Heart Rate Pulse Sensor	Detect irregular or elevated heart rate
IV Flow Rate Control	Rack and Pinion Gear System (Motorized)	Automate flow rate adjustment
IoT Connectivity	ESP32 + Blynk Platform	Real-time remote monitoring, alerts, and control from smartphones

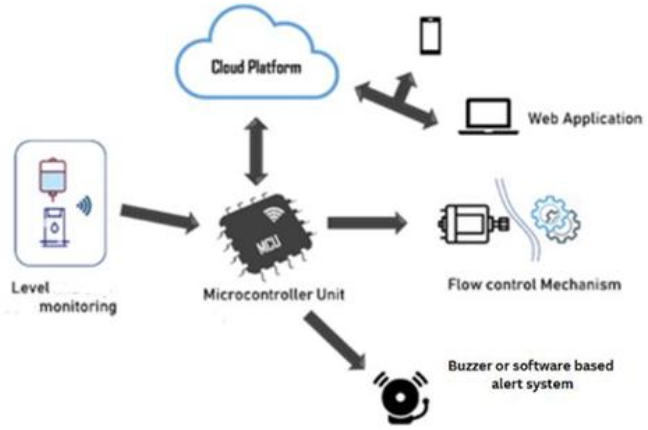
3.3 Circuit Diagram

<u>Overall flowchart of our Circuit:</u>





IOT: We used Blynk for this Purpose.



3.4 Hardware Design

IV Stand: At first we wanted to do a 3D design to customize our stand to accommodate all the specifications that we wanted to achieve like monitoring, controlling and other sensors. But we used cost effective material for hardware prototype.

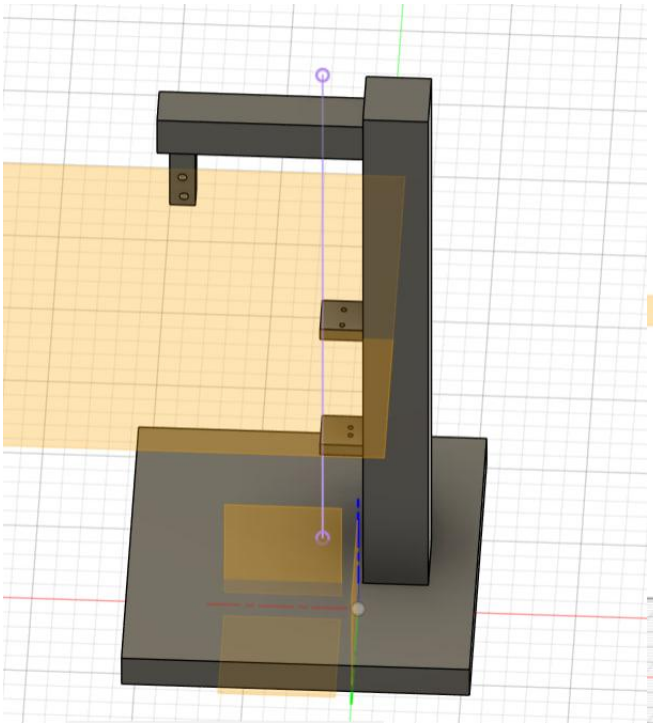


Fig: Stand for Holding IV Bag Only

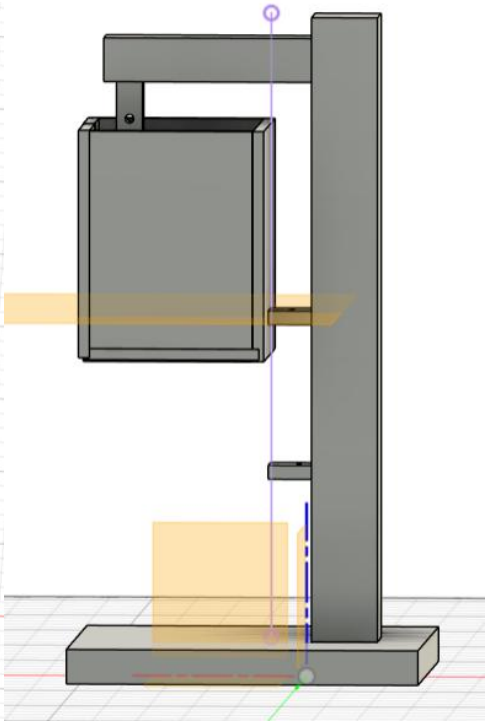


Fig: After Mounting a box for laser Monitoring to avoid light interference

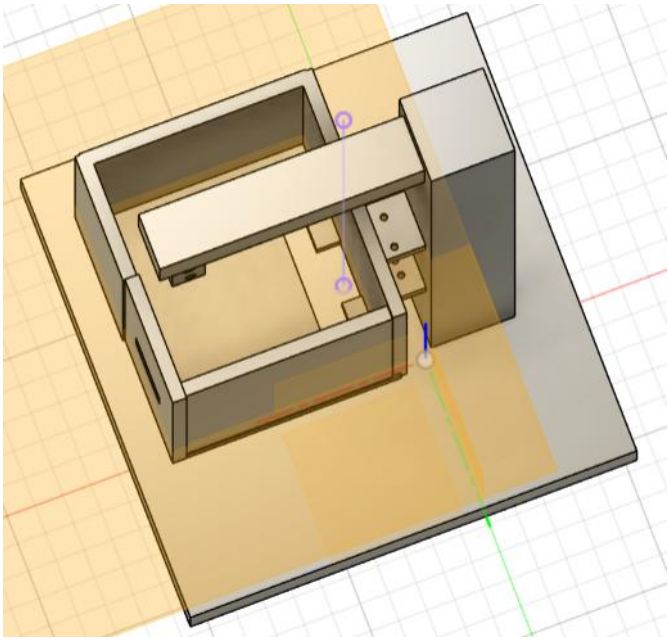


Fig: In Bottom-plane, open space for Drip Chamber

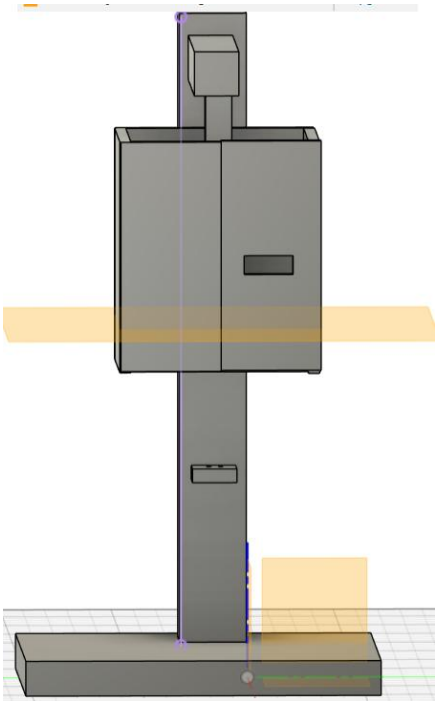


Fig: Door System of the box for Quick and Easy IV Bag Exchange

Rack and Pinion Gear System

This System is successfully implemented by 3D print. This is the key element of our controlling part. Real pictures are shown in the implementation part.

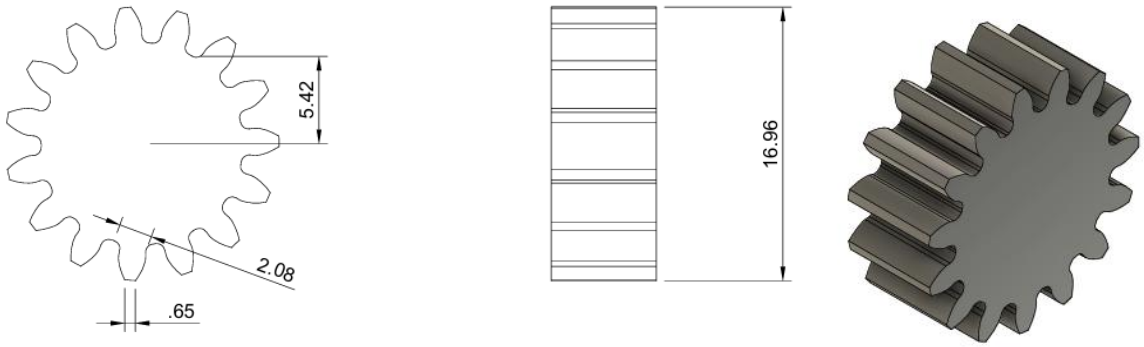


Fig: Pinion Gear System

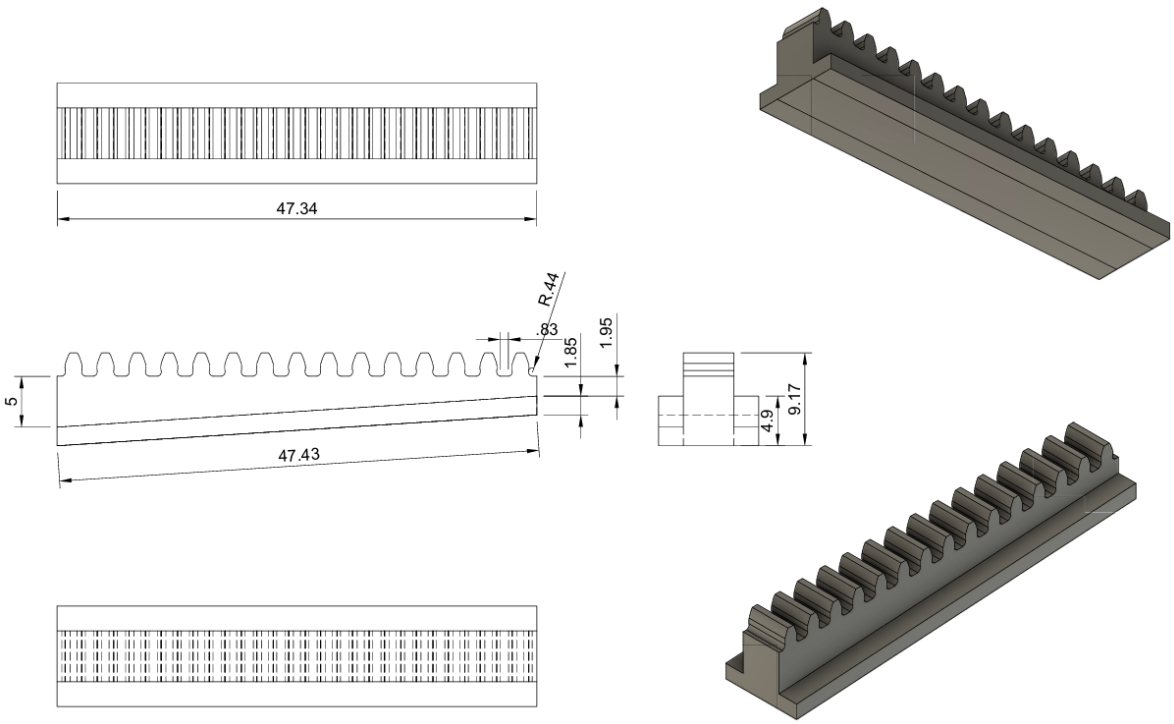


Fig: Rack

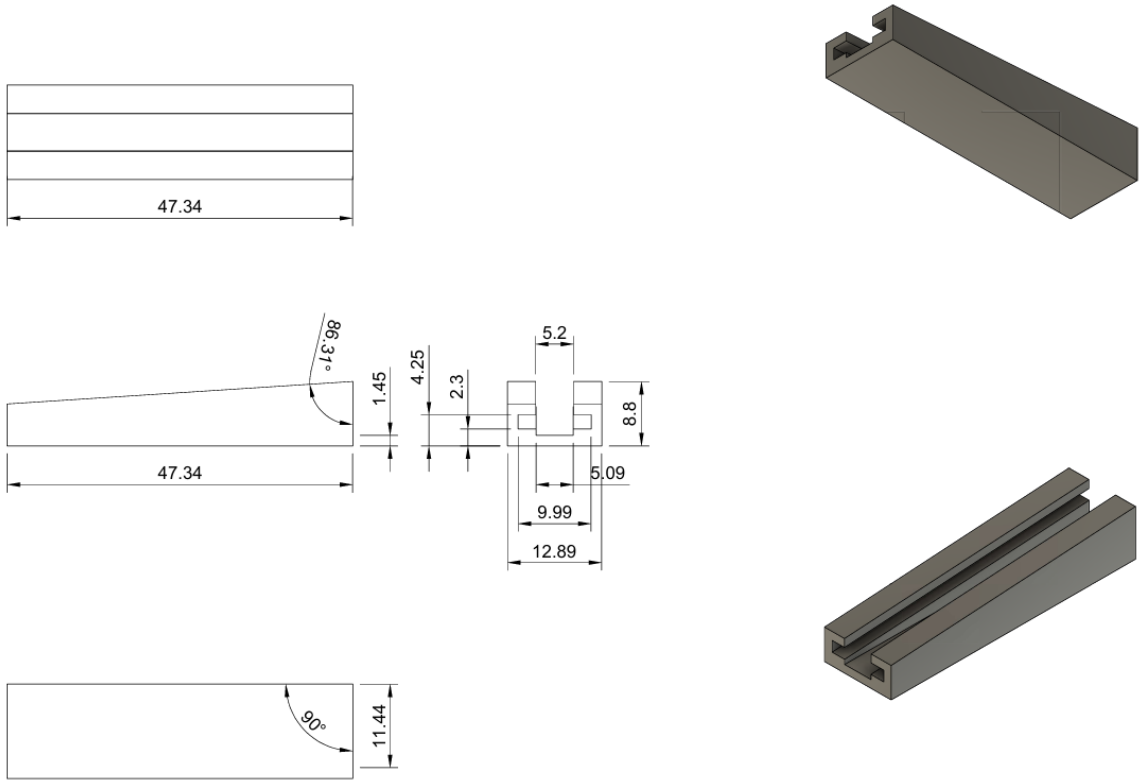


Fig: Holder

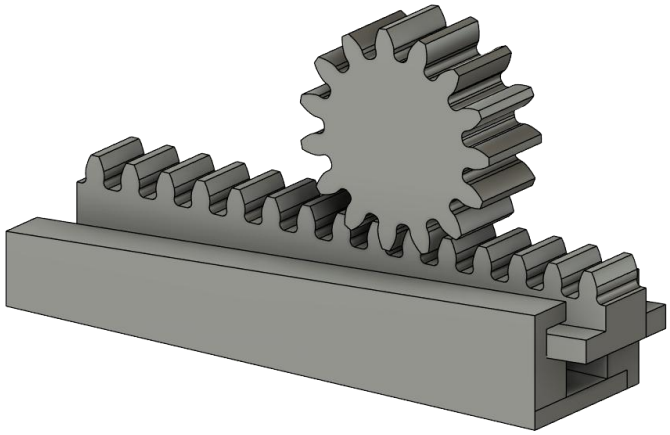


Fig: Final 3D Design

3.5 Full Source Code of Firmware

```
//Arduino and Sensor
// === Pin Definitions ===
#define LDR_PIN A1
#define TEMP_PIN A0
#define PULSE_PIN A2
#define LASER_PIN 7
#define LED_PIN 13

// === Pulse Sensor Constants ===
int threshold = 550;
bool pulseDetected = false;
const int numSamples = 10;
int samples[numSamples];
int sampleIndex = 0;
const int offset = 60;

// === Timing Variables ===
unsigned long lastSensorTime = 0;
int currentSensor = 0;  // 0: LDR, 1: Temp, 2: Pulse
bool laserOn = false;

// === Setup ===
void setup() {
  Serial.begin(9600);
  pinMode(LDR_PIN, INPUT);
  pinMode(TEMP_PIN, INPUT);
  pinMode(PULSE_PIN, INPUT);
  pinMode(LASER_PIN, OUTPUT);
   pinMode(LED_PIN, OUTPUT);
```

```
#define RXp2 16   //ESP32 Code
#define TXp2 17

#define BLYNK_TEMPLATE_ID "TMPL6VehGpQNW"
#define BLYNK_TEMPLATE_NAME "servo"
#define BLYNK_AUTH_TOKEN "0j83EuIorWi4-
oTj4FS1DrYPO2zMRPzQ"

#define BLYNK_PRINT Serial

#include <WiFi.h>
#include <WiFiClient.h>
#include <BlynkSimpleEsp32.h>
#include <ESP32Servo.h>

char ssid[] = "Itachi";
char pass[] = "123345678";

Servo regulatorServo;
int servoPin = 13; // Servo Pin (Pin 13 for
example)
int blueLED = 2;   // LED Pin for control via
Blynk

BlynkTimer timer;
int sensorType = 0; // 0: LDR, 1: Temp, 2: Pulse

// Function to send sensor data to
Blynk
void sendToBlynk(int value) {
```

```
  digitalWrite(LASER_PIN, LOW);

  // === Laser ON ===
  digitalWrite(LASER_PIN, HIGH);
  laserOn = true;
  Serial.println("☑ Laser ON. Monitoring
begins...");
}

// === Main Loop ===
void loop() {
  if (laserOn && millis() - lastSensorTime >=
1000) {
    lastSensorTime = millis();

    switch (currentSensor) {
      case 0:
        sendLDR();
        break;
      case 1:
        sendTemp();
        break;
      case 2:
        sendPulse();
        break;
    }

    currentSensor = (currentSensor + 1) % 3;
  }
}

// === LDR ===
void sendLDR() {
  int light_val = analogRead(LDR_PIN);
  Serial.println(light_val);  // Sent to ESP32
  delay(2000);  // Hold for 2 seconds
}

// === Temperature ===
void sendTemp() {
  int raw = analogRead(TEMP_PIN);
  float voltage = (raw / 1024.0) * 1100.0;
   float tempC = voltage / 10.0;
  Serial.println((int)tempC);  // Sent to ESP32
  delay(2000);  // Hold for 2 seconds
}

// === Pulse Sensor ===
void sendPulse() {
  switch (sensorType) {
    case 0:
      Blynk.virtualWrite(V2,
value);  // Virtual Pin V2 for LDR
data
      break;
    case 1:
      Blynk.virtualWrite(V3,
value);  // Virtual Pin V3 for
Temperature data
      break;
    case 2:
      Blynk.virtualWrite(V4,
value);  // Virtual Pin V4 for Pulse
data
      break;
  }
  sensorType = (sensorType + 1) %
3;  // Move to the next sensor
}

// Blynk virtual button control for
servo (V1)
BLYNK_WRITE(V1) {
  int pos = param.asInt();  // Get
servo position from Blynk app
  regulatorServo.write(pos);  //
Move servo to the specified position
  Serial.print("Servo set to: ");
  Serial.println(pos);
}

// Blynk virtual button control for
LED (V0)
BLYNK_WRITE(V0) {
  int state = param.asInt();  // Get
LED state from Blynk app
  digitalWrite(blueLED, state);  //
Control the LED
  Serial.println(state ? "LED ON" :
"LED OFF");
}

void setup() {
  Serial.begin(115200);
// Start serial communication for
debugging
```

```
  int raw = analogRead(PULSE_PIN);
  samples[sampleIndex] = raw;
  sampleIndex = (sampleIndex + 1) % numSamples;

  long sum = 0;
  for (int i = 0; i < numSamples; i++) sum += samples[i];
  int avg = sum / numSamples;

  float bpm = 0.067 * avg + offset;

  if (avg > threshold && !pulseDetected) {
    pulseDetected = true;
    digitalWrite(LED_PIN, HIGH);
  }

  if (avg < threshold) {
    pulseDetected = false;
    digitalWrite(LED_PIN, LOW);
  }

  Serial.println((int)bpm);  // Sent to ESP32
  delay(2000);  // Hold for 2 seconds
}
```

```
  Serial2.begin(9600, SERIAL_8N1, RXp2, TXp2);  // Start serial communication with Arduino

  pinMode(blueLED, OUTPUT);        // Set the LED pin as output
  regulatorServo.attach(servoPin); // Attach the servo to its pin

  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);  // Connect to Blynk server
}

void loop() {
  Blynk.run();      // Run the Blynk process
  timer.run();      // Run the timer for scheduled tasks

  // Check if data is available from Arduino (via Serial2)
  if (Serial2.available()) {
    String received = Serial2.readStringUntil('\n');  // Read incoming data until newline
    if (received.length()) {
      int value = received.toInt();  // Convert the received data to integer
      Serial.print("Received: ");
      Serial.println(value);  // Debug print the received value
      sendToBlynk(value);     // Send the received value to Blynk
    }
  }
}
```

Table: Source Code for the main program

4 Implementation

4.1 Description

IV Stand:

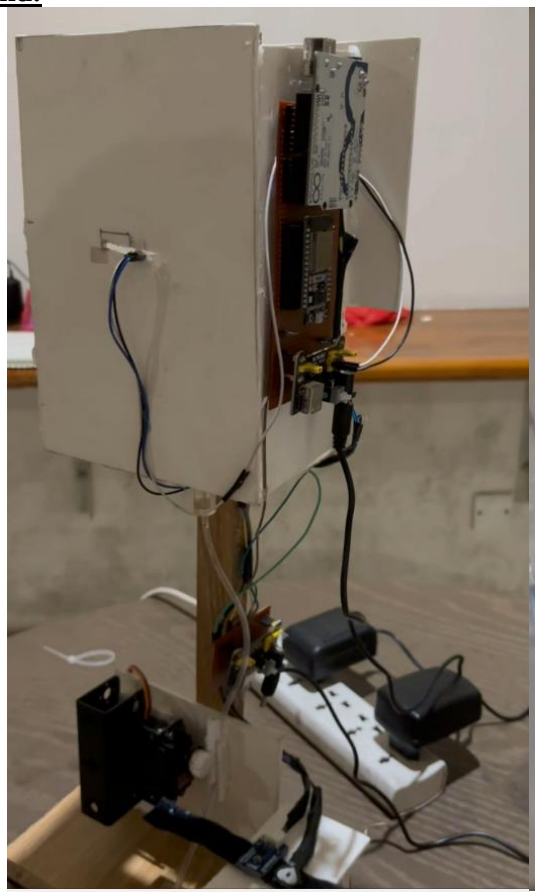


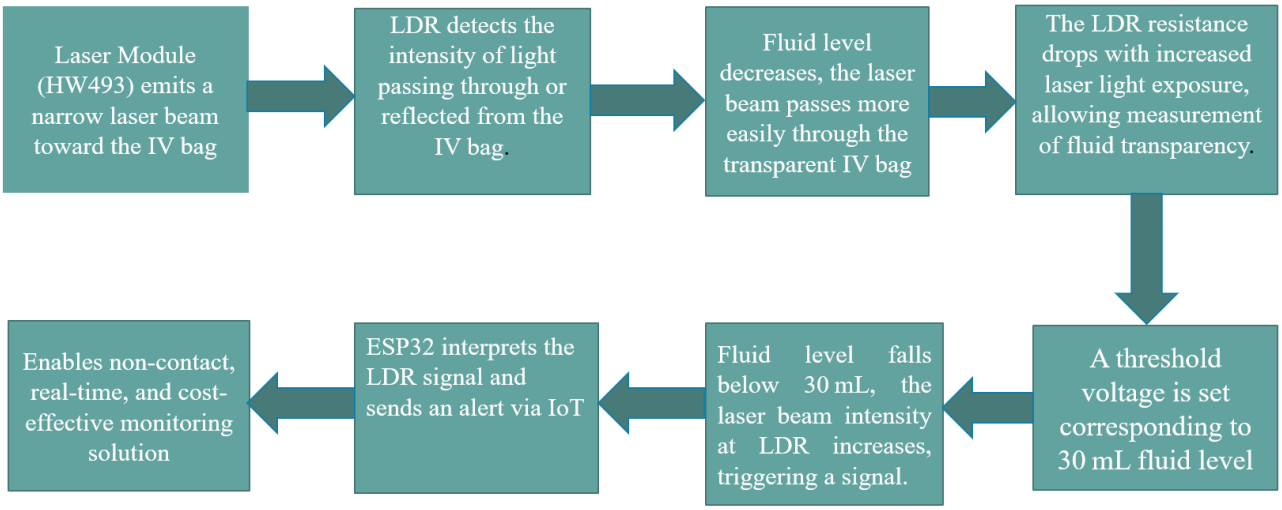
Fig: Front+Side View



Fig: Top View

As mentioned before for cost effectiveness we couldn't 3D print our customized stand. So we had to do it by hand via plywood and PVC.

Laser Monitoring:



Rack And Pinion Gear System: Here PLA is used for 3D printing Purpose.



Proposed rack and pinion mechanism for converting motor's rotational motion into linear motion.

The system consists of :

- a pinion gear-rotating part and
- a toothed rack –sliding part

- ❑ The motor will rotate the pinion gear which in turn causes the rack to slide linearly.
- ❑ The bottom surface of the rack is designed with a slope (incline) that adjusts the pressure applied on the fluid pipe.

Proposed Regulator:

3 selection mode :

-High

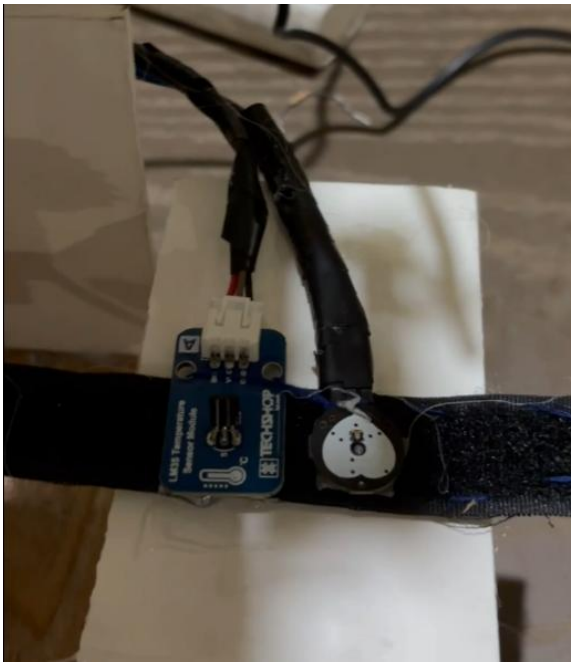
-Medium

-Low

Based on the selected mode (High/Medium/Low), the motor will turn precisely to the required angle, positioning the slope correctly:

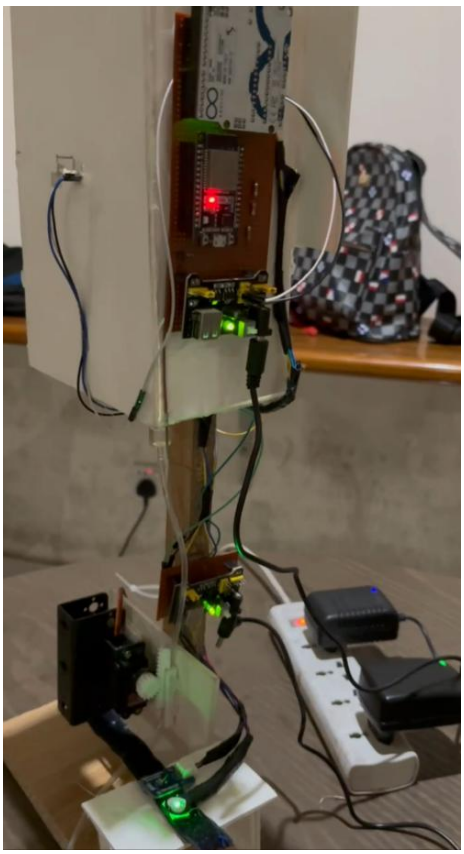
Mode	Degree	Pressure On IV Pipe	Fluid Flow
High	60 to 180 degree	Minimal	Maximum Flow
Medium	60 to 120 degree	Moderate	Moderate Flow
Low	0 to 60 degree	High	Reduced Flow

Wearable Temperature & Pulse Sensor:



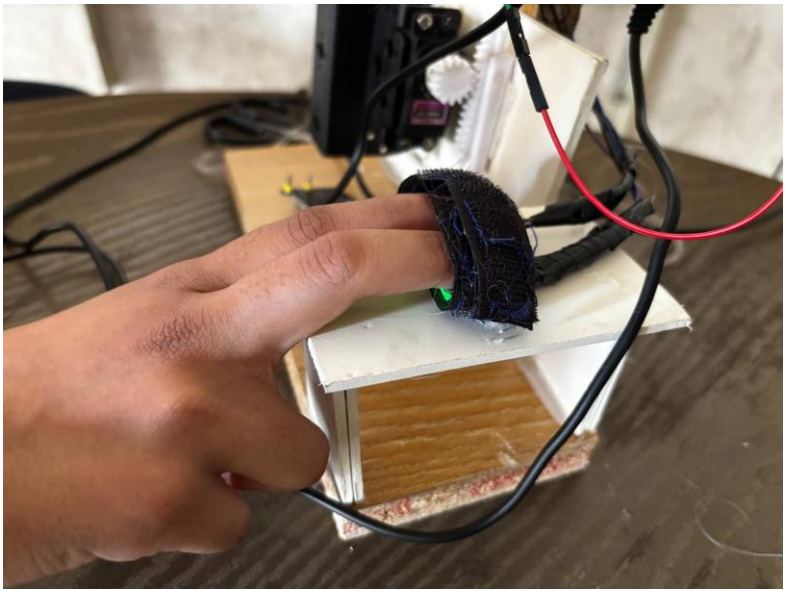
Overall Hardware Prototype:

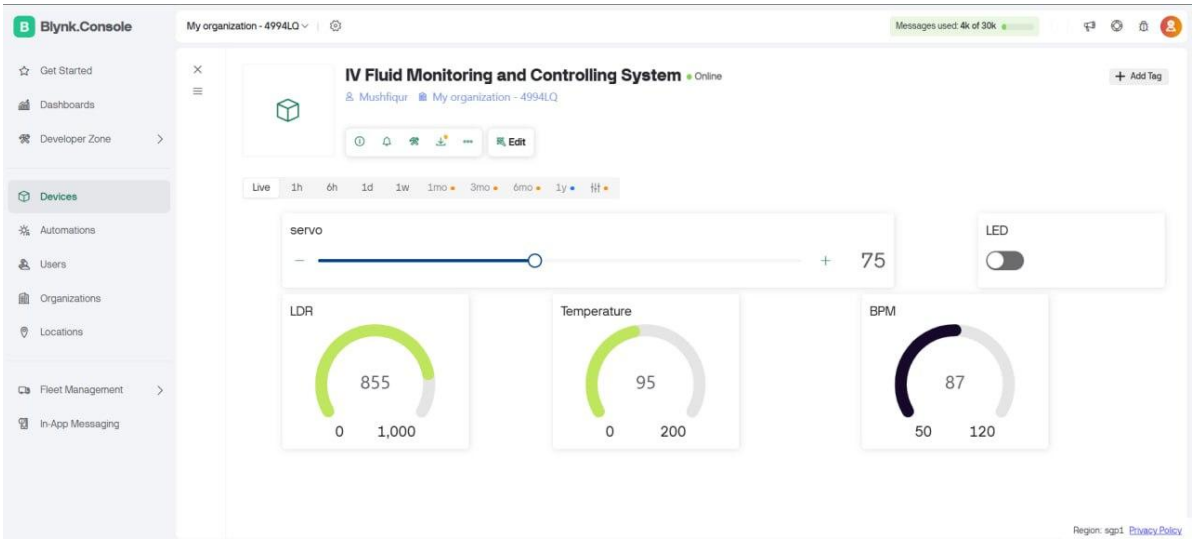
This is the overall setup incorporating laser,temperature and pulse sensors and customized servo motor for controlling purpose.



4.2 Experiment and Data Collection

Temperature & Bpm sensor:





Here temperature is in Farenhite Unit.

LDR Sensor Data:

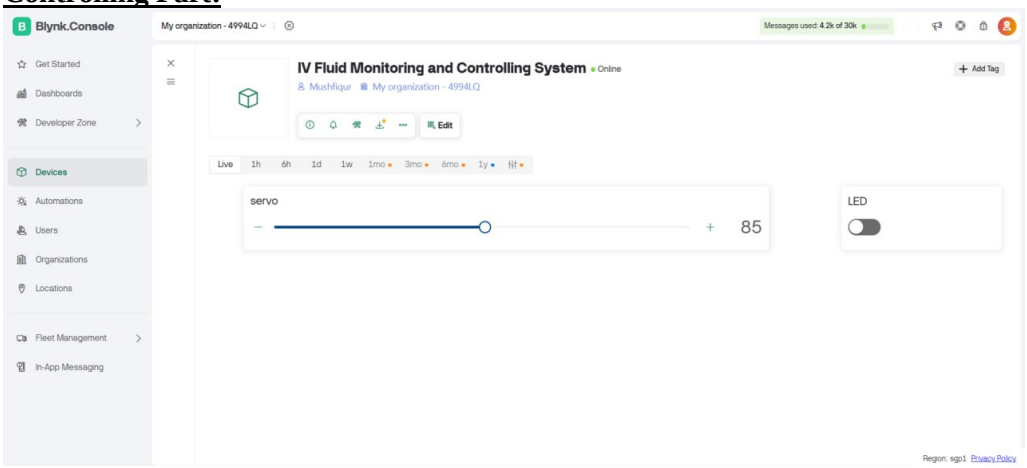
When IV Bag is Full:



When IV Bag is close to 30 mL:



Controlling Part:



4.3 Data Analysis

LDR Sensor: when the IV bag was full of fluid the laser was set such a way that the light incident directly on ldr thats why the reading from ldr was high .
when the bag becomes empty the laser dont incident on the ldr directly , thats why the reading becomes comparatively low

Temperature & Bpm Sensor: When temperature value goes above normal range which is 98.6 degree Fahrenheit, that means patient has fever and when bpm is above the range of 60-100, abnormalities is detected and situation needed to be controlled.

4.4 Results

From the data shown above, we can see that temperature is 95 degree Fahrenheit and bpm is 87 which is between the normal range that means no abnormalities have occurred. LDR value is also fluctuating between 426 (more fluid) and 868 which is decrease of fluid.

4.5 YouTube Link

5 Design Analysis and Evaluation

5.1 Novelty

Dual-Level Monitoring: Fluid + Patient Vitals
Conventional IV setups only monitor fluid level or flow manually. Our system simultaneously tracks fluid levels and vital signs like temperature and heart rate, creating a context-aware monitoring system. This enables int**e**lligent decisions: for instance, adjusting flow rate based on the patient's condition—a feature rarely found in basic automated IV systems.

Hybrid Sensing Approach (Laser + Photodiode)
Many smart IV projects use weight sensors or ultrasonic methods to detect fluid levels. This design uses a laser-photodiode pair, which is a cost-effective, non-contact, and reliable method for tracking fluid depletion. The use of a potentiometer for threshold calibration adds flexibility for different fluid containers.

Smart Mechanical Control Using Rack and Pinion System
Instead of a basic servo or valve, the system uses a rack and pinion gear mechanism for more precise and linear control of IV flow. This offers greater mechanical reliability and fine-tuned flow regulation, enhancing safety and adaptability.

IoT-Enabled Real-Time Control and Alerting
Unlike many systems that only alert on fluid status, this system:

- Sends real-time notifications based on vital signs and fluid levels.
- Allows remote flow rate control through the Blynk IoT platform—making it interactive and fully remote-controllable. Doctors/nurses can monitor multiple patients simultaneously through a centralized dashboard.

Proactive Medical Decision Support
By detecting abnormalities like fever or abnormal pulse, and responding through automated or remote actions, the system acts like a decision-assist tool. This reduces medical response time and improves treatment outcomes.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to public health and safety

Traditional IV therapy systems rely heavily on manual monitoring, which can lead to delayed responses in case of fluid exhaustion or flow obstruction. These situations can cause air embolism, vein collapse, or tissue infiltration—all of which pose serious health risks. Our system mitigates these hazards by: Continuously monitoring IV fluid levels using a laser-photodiode sensor system that detects critical low-fluid thresholds, Automatically adjusting the flow rate via a rack-and-pinion motorized control mechanism, ensuring uninterrupted and optimal delivery of fluids.

The inclusion of body temperature (LM35) and heart rate sensors enables early detection of health anomalies such as fever or tachycardia. These parameters are crucial indicators of infection, dehydration, or adverse drug reactions. Prompt alerts generated by the system help medical staff:

- Intervene quickly to prevent complications and adjust treatment strategies in real time.
- Monitor high-risk patients more effectively, especially in ICUs or emergency rooms.

Manual IV fluid management can be affected by fatigue, multitasking, or inattention by healthcare workers, especially during night shifts or in understaffed facilities. Our IoT-integrated system:

- Sends automated, minimizing reliance on manual checks.

To ensure safety in all scenarios, the system includes: Manual override options to adjust flow rate in case of IoT platform unavailability or device malfunction. Alarm systems that function independently of cloud connectivity to alert staff in critical conditions.

5.2.2 Considerations to environment

Environmental sustainability is an essential consideration in the design of our IoT-Based Intravenous Fluid Monitoring and Flow Control System. The system has been developed with attention to minimizing environmental impact across its life cycle—covering component selection, energy efficiency, and disposal.

The system is powered by an ESP32 microcontroller, which is known for its low power consumption, especially in sleep and idle modes. This makes the device suitable for long-term use without excessive energy demand.Efficient power management helps reduce hospital power consumption when deployed at scale and supports potential battery-powered operation in low-resource settings.

The design utilizes widely available, reusable modules (e.g., sensors, ESP32, motor drivers), which can be easily reprogrammed or repurposed for future projects. By using standard breakout boards and non-custom PCBs for the prototype, we ensure that components are not single-use, which contributes to a reduction in electronic waste (e-waste).

By preventing over-infusion or under-infusion events, the system helps avoid wastage of medical-grade IV fluids, packaging, and tubing—thereby indirectly reducing biomedical plastic waste. The early detection of abnormalities also reduces the need for emergency interventions, which often involve additional disposables like cannulas, syringes, and gloves.

The system avoids the use of hazardous components like mercury-based sensors or leaded solder.

In a mass-deployment scenario, the modular design supports centralized updates (firmware-over-the-air), reducing the need for physical servicing, which in turn minimizes transportation-related carbon emissions. Long-term deployment and durability reduce frequent replacements, contributing to a lower environmental footprint over time.

5.2.3 Considerations to cultural and societal needs

The IoT-Based Real-Time Intravenous Fluid Monitoring and Flow Control System has been carefully designed to respect and address the diverse cultural and societal contexts in which it may be deployed. The following considerations were taken into account:

The system uses cost-effective components like the ESP32, LM35, and photodiode-laser pairs instead of expensive alternatives such as ultrasonic sensors or proprietary medical-grade modules. This affordability ensures that rural clinics, small healthcare centers, and developing regions can

adopt the system without financial strain, contributing to healthcare equity.

In societies where medical professionals are in short supply or where nurse-to-patient ratios are very high, this system can reduce workload through automation, enabling staff to focus on more complex care tasks.

The system only collects medically relevant data (temperature, pulse, fluid level) and transmits it securely using encrypted IoT protocols, ensuring patient data confidentiality and compliance with cultural expectations of privacy. There are no intrusive monitoring mechanisms such as video or audio capture.

By preventing IV-related complications and enabling timely interventions, the system reduces hospitalization time and improves treatment outcomes—benefits that ripple outward to families and communities through improved productivity and reduced medical costs. Its deployment can contribute to broader public health resilience, especially in times of healthcare crises (e.g., pandemics, disaster zones).

5.3 Investigations (PO(d))

5.3.1 Experiment Design

We developed a modular experimental setup and each module was tested independently and then integrated for system-level trials. To verify system functionality, with separate units for:

1. Fluid Level Detection:
    - Tested different thresholds using the laser-photodiode pair and potentiometer.
    - Recorded output voltage against decreasing fluid levels to establish sensitivity curves.
2. Body Temperature and Pulse Monitoring:
    - Validated LM35 and Heart Rate Pulse Sensor
3. Flow Control Mechanism:
    - Designed a rack and pinion system with motor calibration.
    - Assessed flow rate adjustments by observing drip counts before and after actuation.
4. IoT Transmission:
    - Sent real-time sensor data via ESP32 to Blynk dashboard.
    - Simulated emergency conditions to test alarm triggers and remote override.

5.3.2 Data Analysis and Interpretation

5.4 Limitations of Tools (PO(e))

Despite its innovative features, the proposed system has certain limitations stemming from the tools and components used. The accuracy of fluid level detection using a laser-photodiode module may be affected by ambient lighting conditions or inconsistent IV bag materials, potentially leading to false readings. The ESP32 microcontroller, while efficient and cost-effective, relies on stable Wi-Fi connectivity, which may not be consistently available in rural or emergency settings, limiting real-time data transmission. Additionally, the Blynk platform, though user-friendly, has data upload rate restrictions and limited API call frequency in its free version, which could hinder continuous high-frequency monitoring. Furthermore, sensors like the LM35 and basic pulse sensors may lack the precision and robustness required for clinical-grade measurements, affecting reliability in critical care situations. These limitations highlight the need for careful calibration, fallback mechanisms, and potential upgrades in future iterations for more robust and scalable deployment.

5.5 Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

The proposed system offers significant advancements in healthcare accessibility and equity, particularly in resource-limited settings. Its affordability and portability make it ideal for rural and semi-urban clinics, where there is often a shortage of skilled healthcare professionals and a high demand for automation to reduce the need for continuous human monitoring. Additionally, its use of the LM35 temperature sensor, which enables minimal-contact temperature monitoring, ensures gender and cultural sensitivity—especially important in conservative environments where physical interaction between male and female patients or healthcare providers may be restricted. These features collectively enhance the system's adaptability and acceptance across diverse healthcare contexts.

5.5.2 Assessment of Health and Safety Issues

The system is designed to enhance patient safety and clinical efficiency through a combination of proactive, automated, and data-driven features. A real-time alert mechanism notifies medical staff of critical conditions such as low IV fluid levels, abnormal heart rate, or elevated body temperature, enabling timely intervention to prevent severe complications like air embolism, vein collapse, or unnoticed infections. The integration of a rack and pinion mechanism for automated flow regulation significantly reduces the likelihood of human error during manual adjustments. Furthermore, the system continuously monitors and logs vital parameters, supporting data-driven clinical decisions for more personalized and responsive care. To ensure reliability, the system includes manual override options for emergencies and incorporates failsafe alarms that function independently of Wi-Fi connectivity, guaranteeing the delivery of critical alerts even during network disruptions.

5.5.3 Assessment of Legal Issues

To ensure responsible deployment in clinical settings, the system addresses key regulatory and ethical considerations. Data privacy is a primary concern, especially when integrated with hospital information systems or cloud platforms. To mitigate this risk, the system employs secure, encrypted communication protocols aligned with international data protection standards such as GDPR and HIPAA, ensuring the confidentiality of patient information. Compliance with device certification requirements is also a critical factor; the design anticipates adherence to national regulations like the Directorate General of Drug Administration (DGDA) guidelines in Bangladesh, as well as international standards such as IEC 60601 for medical electrical equipment. Additionally, to address liability concerns, the system incorporates multiple layers of redundancy—including manual override functions and real-time alert mechanisms—to ensure safe operation and reduce risks associated with device failure or malfunction.

5.6 Sustainability and Environmental Impact Evaluation (PO(g))

The proposed IV monitoring system incorporates sustainability at every stage of its design and operation, aligning with global environmental and health priorities. It utilizes modular and reusable components such as the ESP32 microcontroller, laser modules, and low-power sensors, allowing for easy upgrades and minimizing the need for complete system replacement. Energy efficiency is achieved through deep-sleep modes supported by the ESP32 and the use of low-power sensors like LM35, making the system viable for solar-powered or off-grid setups common in rural healthcare. The design emphasizes e-waste minimization through repairability—detachable sensors and control boards enable partial replacements.

Environmentally, the system contributes to reduced biomedical waste by preventing complications that would otherwise require additional medical resources and disposable materials. Its durable build and cost-effective scalability support long-term deployment, particularly in underserved regions.

5.7 Ethical Issued (PO(h))

The development of the IV monitoring system was grounded in a strong ethical framework, prioritizing patient safety, data privacy, inclusivity, and professional responsibility. Guided by the principle of "Do no harm," the design minimizes medical errors and enhances reliability through built-in failsafe mechanisms, manual overrides, and local alarms to ensure patient safety even in cases of system failure or connectivity loss. A core objective of the project was to promote equitable healthcare access; the affordable, scalable design is suited for both urban and rural settings, with culturally sensitive features such as non-contact sensing to accommodate diverse norms. Academic integrity and teamwork were upheld throughout, with all technical contributions being original, appropriately cited, and collaboratively developed under a signed honesty agreement. Ethical challenges such as balancing real-time monitoring with data privacy and avoiding over-dependence on automation were carefully mitigated through minimal data collection and the inclusion of manual override options. The team approached the project with a clear understanding of the real-world implications of medical device design, taking extra care in validation, testing, and documentation to uphold the highest standards of professional responsibility.

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

2006160- Temperature+bpm Monitoring+Assembly

2006161- IOT (Blynk) +Integrating All Sensors+ Assembly

2006162- Fluid Monitoring+3D design of Customized Stand+Assembly

2006163- Fluid Control+3D design of customized Gear System+Assembly

6.2 Logbook of Project Implementation

Week	Milestone Achieved
4th Week	Project Idea Finalized + Components Purchased
5th Week	Sensor Module+Motor Tested Successfully
7th Week	3D Design of Customized Stand+Motor
9th Week	Assembly of all the Sensors
12th Week	Flow Control motor assembled
13th Week	Blynk Dashboard Configured and Tested
14th Week	Final Project Demonstration+Report Submission

7 Communication (PO(j))

7.1 Executive Summary

Smart IV System Promises Safer Patient Care with Real-Time Alerts and Remote Monitoring

A group of BUET students has developed a low-cost, IoT-based intravenous (IV) fluid monitoring and control system that enhances hospital safety. The system tracks fluid levels, body temperature, and heart rate in real time and automatically adjusts the IV flow rate based on the patient's condition. Doctors and nurses receive instant alerts on their smartphones when a problem arises, enabling faster intervention and reducing the risk of medical complications. This smart healthcare solution is affordable, contactless, and scalable—especially useful in overburdened or rural healthcare facilities.

7.2 User Manual

System Overview:

This system is designed to monitor IV fluid levels and patient vitals and adjust the flow rate automatically. Data is transmitted to a smartphone dashboard using the Blynk IoT platform, enabling real-time monitoring and control.

Step-by-Step Setup & Usage Guide:

Hardware Setup:

1. **Mount the Sensors:**
    - Attach the laser and photodiode pair across the IV bottle to detect fluid level drops.
    - Place the LM35 near the patient's skin.
    - Attach the pulse sensor to the patient’s fingertip or earlobe.
2. **Connect to ESP32 Board:**
    - Connect all sensors to designated GPIO pins on the ESP32 board.
    - Connect the motorized rack-and-pinion mechanism to control the IV roller clamp.
3. **Power the System:**
    - Use a 5V USB power bank or adapter to supply power to the ESP32 and other components.

Software Setup:

4. **Configure Blynk Dashboard:**
    - Use the Blynk mobile app.
    - Create virtual pins for:
        - Body temperature display
        - Heart rate display
        - Fluid level status
        - Flow rate control (slider or button)
        - Alert notifications
5. **Upload Code:**
    - Flash the firmware (Arduino IDE) onto the ESP32 using a USB cable.
    - Ensure Wi-Fi credentials and Blynk auth token are updated in the code.
6. **System Calibration:**
    - Use the potentiometer to calibrate the light threshold for fluid level detection.
    - Verify sensor readings using a thermometer and pulse oximeter.

Daily Usage:

7. **Start Monitoring:**
    - Once powered on, the system will start monitoring all parameters.
    - Alerts will be pushed to the app if:
        - Fluid level is low
        - Temperature exceeds fever threshold
        - Pulse rate is abnormal
8. **Adjust Flow Rate:**
    - The system will automatically adjust the IV flow rate when needed.
    - Manual override is possible via the app interface (increase or decrease using a slider/button).
9. **Shut Down:**
    - To turn off, simply unplug the system or use the power button on the power source.

8 Project Management and Cost Analysis (PO(k))

8.1 Bill of Materials

Component	Specification / Model	Quantity	Unit Price (BDT)	Total (BDT)
ESP32 Microcontroller, Arduino	NodeMCU ESP32 Dev Board	1+1	1250	1250
Laser Module	HW493	1	250	250
Photodiode	Generic IR Photodiode	1	30	30
Potentiometer	5kΩ	1	20	20
LM35 Temperature Sensor	Analog	1	50	50
Pulse Sensor	Optical sensor	1	370	370
MG996R Servo Motor with Gear (for Rack & Pinion)	6V – 12V	1	300	300
Rack and Pinion Gears	3D printed	1 set	150	150
Jumper Wires, Breadboard,vero board,resistor	–	–	560	560
Power Supply (Adapter)	5V, USB Type	3	1550	1550

→ Total Prototype Cost: ≈ BDT 4160

8.2 Timeline of Project Implementation

Date	Milestone	Team Responsibility
02 May 2025	Project idea finalized	All members
05 May 2025	Component list prepared and procured	Procurement lead
10 May 2025	Sensor modules tested individually	Hardware team

15 May 2025	Integration of sensors with ESP32	Firmware team
20 May 2025	Flow control hardware (rack & pinion) assembled	Mechanical subteam
25 May 2025	Blynk dashboard configured + tested	IoT/Cloud integration team
28 May 2025	Prototype debugging + calibration completed	Whole team
01 June 2025	Data collected from simulation and testing	Experimentation and analysis lead
03 June 2025	Final presentation prepared	Documentation + communication subteam
04 June 2025	Project report submitted	All members

9 Future Work (PO(l))

1. Enable seamless integration with existing hospital databases and Electronic Health Records (EHR). Automatically log patient vitals and IV history for medical staff review and long-term tracking.
2. Implement machine learning algorithms to predict optimal IV flow rates based on patient data trends. Detect early signs of medical emergencies (e.g., sepsis, dehydration) from sensor data. Perform anomaly detection for flow inconsistencies or sensor faults.
3. Expand the system into a multi-node network to monitor several patients wirelessly from a central station. Assign individual IDs for each system, enabling grouped or ward-level patient tracking.
4. Include voice-controlled interfaces (e.g., via Google Assistant or Alexa) for hands-free operation by nurses/doctors. Design an assistive mobile interface for visually or physically impaired medical staff.
5. Add sensors for oxygen saturation (SpO_2)**,** blood pressure, body movement detection to monitor patient mobility or agitation.

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